Classifying Optical Appearance beyond the Band Gap: Effective Conduction Dimensionality and Electronic Delocalization in Carbon Allotropes

K. Miyauchi
Independent Researcher
(Dated: April 5, 2026)

The macroscopic optical appearance of materials—transparent, colored, black, or lustrous—cannot be fully classified by the band gap alone. Both metals and graphite possess a vanishing band gap, yet the former exhibits metallic luster while the latter appears black with cleavage-plane gloss.

We introduce two classification variables—an effective delocalization index δ and an effective conduction dimensionality D_{eff}—and test them on five carbon allotropes (diamond, C_{60}, SWCNT, graphene, graphite) with hexagonal boron nitride (h-BN) as a control. A decision tree based on E_g and D_{eff} correctly classifies 6 of 7 materials (85.7%) into the correct optical-response category, with the sole failure—single-layer graphene—requiring the layer count N as a third variable.

As supporting evidence, we compute the full tight-binding Hamiltonian $\rightarrow$ Kubo formula $\rightarrow \varepsilon(\omega)$ $\rightarrow$ Fresnel reflectivity chain for each system and test disorder robustness via Anderson disorder simulations. Systems with high $\delta \times D_{\text{eff}}$ maintain their optical response under disorder, while low-$\delta \times D_{\text{eff}}$ systems rapidly decohere.

The framework does not replace band-gap theory but extends it: E_g determines *at which energy* the response occurs, while D_{eff} determines its *directionality* and δ its *robustness*. Extension to liquid and interfacial systems is proposed as a future target.

I. INTRODUCTION

A. The luster question

Both metals and graphite possess a vanishing band gap, yet the former exhibits metallic luster while the latter appears black. Conventional physics explains metallic luster via the Drude model and dielectric reflection via the Fresnel equations, but no unified framework classifies the full range of optical appearances—transparent, colored, black, and lustrous—from a common set of variables.

B. Particle freedom as the classification principle

We propose that the relevant classification variable is the *freedom of constituent particles*, quantified by a delocalization index δ:

- **Localized particles** (δ low): each scatterer responds independently; re-emitted phases are randomized. Result: diffuse scattering (transparent or white).
- **Delocalized particles** (δ high): scatterers respond collectively; re-emitted phases are aligned. Result: specular reflection (luster).

C. Scope: proof of concept in solid carbon

The present work is a **proof of concept**: we test this idea in the most controlled setting available. In solids, nuclear degrees of freedom are frozen, isolating the effect of electronic delocalization δ_{elec}. Carbon allotropes are chosen because a single element spans the full range of optical appearances—transparent (diamond), colored (C_{60}, semiconducting SWCNTs), black (graphite), and lustrous (graphite cleavage plane)—allowing us to vary δ and D_{eff} while eliminating elemental composition as a confounding factor.

D. Gap in the existing classification

The macroscopic optical appearance of materials falls into four broad categories: transparent (diamond), colored (ruby), black (graphite), and metallic luster (silver). Existing theories explain each category individually, but no unified predictive framework based on a common set of classification variables has been established.

In particular, the following questions lack a unified answer:

- Why do metals and graphite, both having a vanishing band gap, differ so dramatically in optical response?
- Why are h-BN and graphite—both layered sp^2 networks—white and black, respectively?
- Why does a single element (carbon) cover all four optical categories?

E. From classification to prediction

Previous attempts at optical classification have been descriptive: given a known material, assign it to a category. This work goes further by establishing a *predictive chain* from microscopic quantities (hopping integrals, orbital symmetry) to a macroscopically measurable quantity (gloss in industry-standard units [GU]).

The key advance is the recognition that the Fresnel equations alone do not predict gloss: two materials may have similar $\varepsilon(\omega)$ yet exhibit very different gloss because of their differing robustness to atomic-scale disorder. The $\delta \times D_{\text{eff}}$ framework captures this robustness and thus provides predictions beyond what Fresnel optics can offer.

II. FRAMEWORK

A. Causal structure

At the most general level, the freedom of constituent particles (δ) determines whether an interface acts as a coherent or incoherent quantum barrier for photons. In solid-state electronic systems, this reduces to the product of two variables:

$$\text{Optical response} = f(\delta_{\text{elec}},\ D_{\text{eff}})\,, \tag{1}$$

where δ_{elec} quantifies the degree of electronic delocalization and D_{eff} quantifies the effective dimensionality of electronic conduction. The concept of particle freedom (δ) motivated the introduction of D_{eff} as a classification variable: if the optical response depends on how freely particles can move, one must specify not only *whether* they can move (encoded in E_g) but also *in how many directions* (encoded in D_{eff}).

The central insight is that E_g alone provides an incomplete mapping from electronic structure to optical response. Adding D_{eff} completes the mapping. The measurable classification variables are E_g and D_{eff}; δ serves as the physical intuition that unifies them and whose independent verification in liquid and interfacial systems is proposed as a future target.

B. Effective delocalization index δ: operational definition

δ is not a single physical quantity but an overarching variable characterising electronic delocalization, whose existence is inferred from the mutual consistency of several proxy indicators. As we discuss in Sec. V F, the maximally localized Wannier function spread Ω [1] provides a rigorous first-principles realisation of δ; the proxies below serve to establish its relevance without requiring a Wannier calculation:

TABLE I. Candidate proxy indicators for δ.

Indicator	Definition	Advantage
π-band width W_π	Full energy width of the π band (eV)	Direct comparison among sp^2 systems
Inverse effective mass $1/m^*$	Curvature of $E(k)$ at band edge	Defined for all materials
IPR	$\sum_i \lvert\psi_i\rvert^4$	Direct measure of delocalization
Wannier spread	$\langle r^2\rangle - \langle r\rangle^2$	Physically transparent

Our strategy is to show that multiple proxies yield the same ordering across materials, thereby supporting the existence of an overarching variable δ without requiring a unique definition.

Among these proxies, the inverse participation ratio (IPR) serves as the **primary operational definition** because it is directly computable from any Hamiltonian (tight-binding or first-principles) without requiring system-specific quantities such as π-band identification or effective-mass fitting. We define:

$$\delta_{\text{IPR}} = \frac{1}{N_{\text{orb}} \cdot \langle \text{IPR}\rangle_{\text{occ}}}, \quad \text{IPR}_n = \sum_i |\psi_{ni}|^4, \tag{2}$$

where N_{orb} is the number of orbitals and the average runs over all occupied eigenstates. A perfectly delocalized state gives $\text{IPR} = 1/N_{\text{orb}}$ and $\delta_{\text{IPR}} = 1$; a fully localized state gives $\text{IPR} = 1$ and $\delta_{\text{IPR}} = 1/N_{\text{orb}} \to 0$. The remaining proxies (W_π, $1/m^*$) serve as supporting evidence that the ordering is robust across different measurement approaches.

C. Effective conduction dimensionality D_{eff}: operational definition

D_{eff} is defined as the **effective rank of the group-velocity tensor** evaluated for electronic states near the Fermi level or band edges:

$$D_{\text{eff}} = \frac{\text{Tr}(\mathbf{T})^2}{\text{Tr}(\mathbf{T}^2)}, \quad T_{\alpha\beta} = \frac{1}{N_k}\sum_{\boldsymbol{k}} \sum_{n\in\text{occ}} \sum_{m\in\text{unocc}} v^{\alpha}_{nm}\, v^{\beta *}_{nm}, \tag{3}$$

where $v^{\alpha}_{nm} = \langle n\boldsymbol{k}|\partial_\alpha H|m\boldsymbol{k}\rangle$ is the interband velocity matrix element. This definition yields $D_{\text{eff}} = d$ for a system with d equivalent conduction directions and $D_{\text{eff}} = 0$ for a system with no band dispersion (molecular solid).

This definition is *purely electronic and structural*: it depends only on the band dispersion $E(\boldsymbol{k})$ and therefore avoids circularity with the optical response it is used to classify.

Energy windowing and the h-BN case. The sum in Eq. 3 runs over *all* occupied-to-unoccupied interband transitions, without restriction to a particular energy

window. For h-BN, the π-band dispersion is substantial ($W_\pi \approx 2.4$ eV), but the band gap $E_g = 5.95$ eV places all interband transitions in the deep UV. The resulting velocity matrix elements v_{nm} are nonzero, and $T_{\alpha\beta}$ reflects the in-plane anisotropy. However, none of these transitions contribute to the *visible-range* optical response, which is the target of our classification.

We acknowledge that D_{eff} and E_g are not fully independent: a large E_g reduces the density of visible-energy transitions, indirectly suppressing the optical weight that enters $T_{\alpha\beta}$. The key point is that D_{eff} provides *additional* information beyond E_g: among materials with the same $E_g \approx 0$ (graphite, metals, SWCNTs), D_{eff} correctly resolves the luster degeneracy that E_g alone cannot lift. The partial correlation with E_g is a feature, not a bug—it reflects the physical reality that both variables derive from the same band structure.

Limitation: frequency-resolved $D_{\text{eff}}(\omega)$. A more rigorous formulation would restrict the sum in Eq. 3 to transitions within an energy window $[\omega - \Delta\omega, \omega + \Delta\omega]$, yielding a frequency-resolved $D_{\text{eff}}(\omega)$ that tracks which spatial directions contribute to the optical response at each photon energy. The present work uses the all-energy definition for simplicity and because it is sufficient to classify the systems studied here. However, for materials where qualitatively different transitions dominate in different spectral ranges (e.g., interband d-to-s vs. intraband Drude in transition metals), a frequency-resolved $D_{\text{eff}}(\omega)$ would be necessary. This extension is left for future work.

TABLE II. Crystallographic vs. effective conduction dimensionality.

Material	Crystal dim.	D_{eff}	Rationale
Diamond	3D	0	$E_g = 5.47$ eV [2]; no states in visible window
C_{60} fcc	3D	0	Intermolecular $t \sim 30$ meV [3]; $W \approx 0.5$ eV [4]
SWCNT	1D	1	Axial dispersion only
Graphene/ Graphite	2D/3D	2	In-plane $W \approx 9$ eV [5] $\gg$ interlayer $\gamma_1 \approx 0.4$ eV
Metals	3D	3	Isotropic conduction
h-BN	2D	0	$E_g = 5.95$ eV [6]; no states in visible window

Note that h-BN possesses a 2D crystal structure with in-plane band dispersion ($W_\pi \approx 2.4$ eV), but because its band gap (5.95 eV) places all π-band states far outside the visible-energy window, $D_{\text{eff}} = 0$. The distinction between crystallographic dimensionality and D_{eff} is essential to the framework.

D. Decision tree for optical response classification

The two variables E_g and D_{eff} define a decision tree:

1. $E_g > E_{\text{vis}} \approx 3.1$ eV?
   → **Yes**: Transparent (no electronic response in the visible range).
   → **No**: proceed to step 2.

2. $E_g > 0$?
   → **Yes**: Colored (selective absorption).
   → **No** ($E_g \approx 0$): proceed to step 3.

3. Classification by D_{eff}:
   $D_{\text{eff}} = 1$: Black with weak luster (1D metallic).
   $D_{\text{eff}} = 2$: Black with cleavage-plane gloss (2D metallic).
   $D_{\text{eff}} = 3$: Full metallic luster (3D metallic).

E. Second-order corrections

The two-variable framework captures the dominant trends. Known second-order effects include:

- **Polarity/ionicity** (p): contributes to h-BN whiteness via π-electron localization.

- **Excitonic and local-field effects**: bound excitons modify optical absorption near the band edge. These effects are particularly large in h-BN (exciton binding energy ∼0.7 eV [7]) and C_{60} (Frenkel excitons). The present tight-binding optical models do not include excitonic or local-field corrections; classification outcomes for these systems should therefore be interpreted as first-approximation results. Including the Bethe–Salpeter equation (BSE) in the optical modeling is a natural next step, though we note that the decision-tree classification relies on E_g and D_{eff} (not on the detailed spectral shape), which are less sensitive to excitonic renormalization than $\varepsilon(\omega)$ itself.

- **Thickness/stacking number** N: the graphene-to-graphite crossover (Beer–Lambert regime).

III. METHODS

A. Choice of benchmark systems

- **Primary**: Five carbon allotropes—diamond, C_{60} (fcc solid), SWCNT (semiconducting and metallic), single-layer graphene, and graphite.

- **Control**: h-BN (isostructural to graphite; same D_{cryst} but low δ).

- **Rationale**: Varying structure within a single element isolates δ and D_{eff} from compositional effects.

B. Collection of δ-proxy data from published sources

The δ-proxy indicators (W_π, $1/m^*$, E_g) in Tables III–**??** were obtained from published literature and open databases. Separately, first-principles DFT + Wannier90 calculations for diamond, graphite, silicon, and germanium were performed as an independent confirmation of the δ ordering; these are described in Sec. V F. The main classification analysis (Secs. IV–**??**) relies exclusively on literature-derived data and tight-binding models.

1. *π-band width W_π*

TABLE III. π-band width data.

Material	W_π (eV)	Source
C_{60} solid	0.4–0.5	[4]
SWCNT	8–9	[5]
Graphene	∼9	TB: $t_0 = 2.7$ eV [8]
Graphite	∼9	[5]
h-BN	∼2.4	[9]

2. *Inverse effective mass $1/m^*$*

TABLE IV. Inverse effective mass data.

Material	m^*/m_e ($1/m^*$)	Source
Diamond	0.48 (2.08)	[10]
C_{60}	4.0 (0.25)	[4]
SWCNT	0.07 (14.3)	[11]
Graphene	0.03 (33.3)	[12]
Graphite	0.039 (25.6)	[13]
h-BN	0.26 (3.85)	[10]

3. *Band gap E_g*

Diamond: 5.47 eV [2]; C_{60}: 1.75 eV [14]; SWCNT (semiconducting): ∼1.0 eV [8]; graphene and graphite: 0 eV; h-BN: 5.95 eV [6].

C. Tight-binding simulation with optical conductivity

Tight-binding models were constructed for four systems (diamond, 1D chain as SWCNT proxy, graphene, and C_{60} cluster). For diamond and graphene, full Slater–Koster models (s, p_x, p_y, p_z; 8 bands per unit cell) were used to capture both σ- and π-band contributions to the optical response. The Kubo formula was used to compute the optical conductivity $\sigma(\omega)$ from the velocity matrix elements $v^\alpha_{nm} = \langle n\boldsymbol{k}|\partial_\alpha H|m\boldsymbol{k}\rangle$, yielding $\varepsilon_2(\omega)$. The real part $\varepsilon_1(\omega)$ is obtained via the Kramers–Kronig transform:

$$\varepsilon_1(\omega) = 1 + \frac{2}{\pi}\mathcal{P}\int_0^\infty \frac{\omega'\varepsilon_2(\omega')}{\omega'^2 - \omega^2}\, d\omega'. \tag{4}$$

For diamond, a scissors correction ($\Delta = 2.87$ eV) shifts the interband onset to the experimental gap $E_g = 5.47$ eV [2], and the high-frequency dielectric constant $\varepsilon_\infty = 5.7$ [15] is imposed to account for electronic transitions beyond the 8-band basis. The normal-incidence reflectivity follows from $R = |(\tilde{n} - 1)/(\tilde{n} + 1)|^2$ with $\tilde{n} = \sqrt{\varepsilon_1 + i\varepsilon_2}$.

The same velocity matrix elements define the group-velocity tensor $\langle v_\alpha v_\beta\rangle$, from which D_{eff} is computed as the effective rank (Eq. 3). The simulation thus derives D_{eff} and $\varepsilon(\omega)$ from the *same* band structure, confirming their mutual consistency.

D. Angular Fresnel reflectivity and gloss prediction

From $\varepsilon(\omega) = \varepsilon_1 + i\varepsilon_2$, the complex refractive index $\tilde{n} = \sqrt{\varepsilon}$ is obtained, and the angular Fresnel reflectivities for s- and p-polarizations are computed:

$$r_s = \frac{\cos\theta_i - \tilde{n}\cos\theta_t}{\cos\theta_i + \tilde{n}\cos\theta_t}, \tag{5}$$

$$r_p = \frac{\tilde{n}\cos\theta_i - \cos\theta_t}{\tilde{n}\cos\theta_i + \cos\theta_t}, \tag{6}$$

where θ_t is determined by Snell's law. The unpolarized reflectivity $R(\theta) = \frac{1}{2}(|r_s|^2 + |r_p|^2)$ is averaged over the visible range (1.6–3.1 eV) to obtain $R_{\text{vis}}(\theta)$.

Gloss is predicted in industry-standard Gloss Units (GU) following ASTM D523 [16]:

$$\text{GU}(\theta) = \frac{R_{\text{vis}}(\theta)}{R_{\text{ref}}(\theta)} \times 100, \tag{7}$$

where R_{ref} is the reflectivity of the black glass reference ($n = 1.567$) [16] at the same angle. Standard measurement geometries are $\theta = 20°$, $60°$, and $85°$.

E. Coherence preservation test: Anderson disorder

To directly test the hypothesis that $\delta \times D_{\text{eff}}$ determines coherence preservation under realistic conditions, we introduce Anderson disorder into the tight-binding Hamiltonian:

$$H \to H + \sum_i \epsilon_i |i\rangle\langle i|, \quad \epsilon_i \in [-W/2, +W/2], \tag{8}$$

where W is the disorder strength. For each system (graphene supercell, 1D chain, C_{60} molecule), we compute:

1. The ensemble-averaged optical conductivity $\langle\sigma(\omega)\rangle$ over 15 realizations.
2. The spectral correlation $\mathrm{corr}(\sigma_0, \sigma_W)$ between the clean ($W = 0$) and disordered spectra.
3. The inverse participation ratio (IPR) as a function of W.

The spectral correlation is defined as the Pearson correlation coefficient between the clean and disordered optical conductivity spectra over the visible range (1.6–3.1 eV, flat spectral weight, $N_\omega = 500$ grid points):

$$\mathrm{corr}(\sigma_0, \sigma_W) = \frac{\sum_i (\sigma_0^i - \bar{\sigma}_0)(\sigma_W^i - \bar{\sigma}_W)}{\sqrt{\sum_i (\sigma_0^i - \bar{\sigma}_0)^2 \sum_i (\sigma_W^i - \bar{\sigma}_W)^2}}. \tag{9}$$

A value near unity means the spectral shape is preserved (coherence maintained); corr $\ll 1$ means spectral features are destroyed (coherence lost).

F. Effective gloss under disorder

The *effective gloss* of a real material with atomic-scale disorder is estimated as:

$$\mathrm{GU}_{\mathrm{eff}} = \mathrm{GU}_{\mathrm{clean}} \times \mathrm{corr}(\sigma_0, \sigma_W)^2, \tag{10}$$

The corr^2 factor is motivated by analogy to two established results: (i) In the Debye–Waller theory of X-ray diffraction, the coherent (Bragg) intensity scales as $I \propto e^{-2W}$, where the exponent encodes the mean-square displacement from ideal positions. Similarly, spectral correlation measures how closely the disordered optical response matches the ideal one, and the specular intensity depends on the *square* of the coherent amplitude fraction. (ii) In radar scattering theory (Beckmann–Spizzichino), the specular component of reflected power from a rough surface scales as $\exp[-(4\pi\sigma_h \cos\theta/\lambda)^2]$—again a squared exponential in the deviation measure. Our corr^2 is the simplest ansatz sharing this quadratic structure.

We emphasize that this is a *minimal phenomenological model*, not a rigorous derivation from surface-scattering theory. The Beckmann–Spizzichino framework requires explicit surface roughness statistics (RMS height σ_h, correlation length τ) that are not modeled here; connecting $\delta \times D_{\mathrm{eff}}$ to these parameters is a concrete target for future work. Nevertheless, Eq. 10 yields a testable prediction: two materials with identical clean-surface reflectivity may have different effective gloss if their disorder robustness differs.

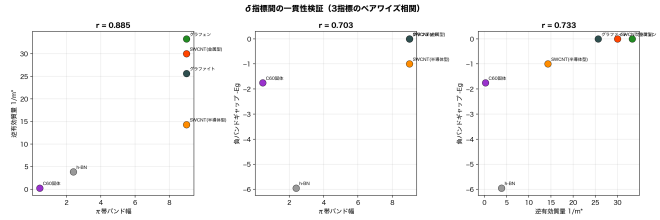

FIG. 1. Pairwise scatter plots among the three δ-proxy indicators, with Pearson and Spearman correlation coefficients.

G. Statistical tests

Pairwise Pearson, Spearman, and Kendall correlations were computed among the three δ-proxy indicators. Classification accuracy of the $E_g \times D_{\mathrm{eff}}$ decision tree was evaluated against observed optical-response categories.

IV. RESULTS

A. Consistency among δ-proxy indicators

TABLE V. Pairwise correlations among three δ-proxy indicators ($N = 5$–6 materials).

Pair	Pearson r	Spearman ρ	Kendall τ	$p(\tau)$
W_π vs. $1/m^*$	0.885	0.845	0.775	0.042
W_π vs. $-E_g$	0.703	0.826	0.674	0.093
$1/m^*$ vs. $-E_g$	0.733	0.880	0.745	0.044

Two of three pairs reach $p < 0.05$ for Kendall's τ, and all three Spearman coefficients exceed 0.82. The consistent ordering across independent indicators supports the existence of an overarching delocalization variable δ.

B. $E_g \times D_{\mathrm{eff}}$ optical-response map

Figure 2 shows all seven materials plotted in E_g–D_{eff} space, with the decision-tree classification boundaries overlaid. Six of seven materials fall into the correct optical-response region; single-layer graphene is the exception (see Table VI).

C. Classification accuracy

Classification accuracy with two variables (E_g, D_{eff}): **6/7 (85.7%)**.

†The sole misclassification: single-layer graphene ($E_g = 0$, $D_{\mathrm{eff}} = 2$) is predicted as "Black+gloss" by the decision tree, but is in fact nearly transparent ($\pi\alpha \approx 2.3\%$ absorption per layer [17]). This failure

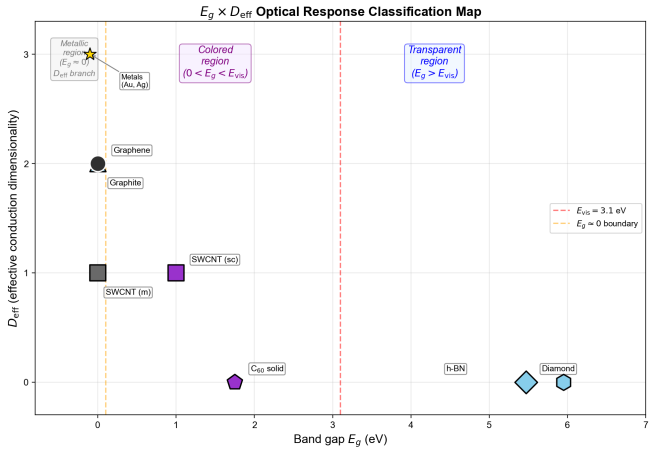

FIG. 2. Optical-response classification map in E_g–D_{eff} space. Shaded regions correspond to decision-tree categories (Sec. II D). Six of seven materials are classified correctly; single-layer graphene ($E_g = 0$, $D_{\text{eff}} = 2$) is misclassified as "Black+gloss" (see Table VI).

TABLE VI. Decision-tree classification results.

Material	E_g (eV)	D_{eff}	Predicted	Observed	
Diamond	5.47	0	Transparent	Transparent	✓
C_{60}	1.75	0	Colored	Colored	✓
SWCNT (sc)	1.0	1	Colored	Colored	✓
SWCNT (m)	0	1	Black−luster	Black−luster	✓
Graphene	0	2	Black+gloss†	Transparent	×
Graphite	0	2	Black+gloss	Black+gloss	✓
h-BN	5.95	0	Transparent	Transparent	✓

reflects a genuine limitation of the two-variable framework: the layer count N acts as a third variable governing the graphene-to-graphite crossover via the Beer–Lambert law. Including N restores 7/7 accuracy but at the cost of an additional parameter.

D. h-BN as a control system

h-BN shares the layered sp^2 network topology of graphite, so one might expect $D_{\text{eff}} = 2$. However, the electronegativity difference between B and N breaks the sublattice (spatial-inversion) symmetry: the π electrons localize on nitrogen sites, reducing W_π from 9 eV (graphite) to 2.4 eV [9] and opening a gap of 5.95 eV [6]. In the language of the present framework, the symmetry-breaking potential localizes the π electrons, *lowering* δ. Since E_g far exceeds the visible-energy window, $D_{\text{eff}} = 0$ despite the 2D crystal structure.

This demonstrates two key points: (i) D_{eff} is not reducible to crystallographic dimensionality, and (ii) δ is an independent variable whose variation can flip the optical response from black to transparent within the same structural family.

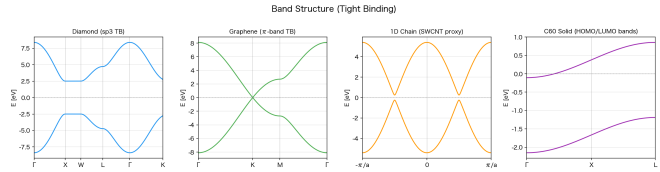

FIG. 3. Tight-binding band structures for four model systems: diamond, graphene, 1D chain (SWCNT proxy), and C_{60} cluster.

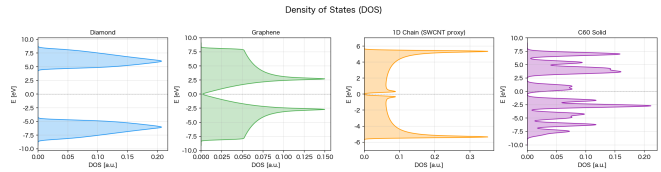

FIG. 4. Density of states for the four model systems. Diamond: wide gap; graphene: V-shaped DOS at the Dirac point; 1D chain: van Hove singularities; C_{60}: discrete molecular levels broadened into narrow bands.

E. Tight-binding simulation results

1. D_{eff} from the velocity tensor

The effective rank of $T_{\alpha\beta}$ (Eq. 3) was computed for each system from the full Slater–Koster (diamond, graphene) or π-band (1D chain) Hamiltonian. Computing over *all* interband transitions yields: diamond: 3.00, graphene: 2.00, 1D chain: 1.00, C_{60}: 0.00—reproducing the expected integer values exactly.

Important distinction: The all-energy D_{eff} reported above measures the *intrinsic structural dimensionality* of the conduction channels. For diamond, $D_{\text{eff}}^{\text{all}} = 3.00$ confirms that interband velocity matrix elements span all three spatial directions. However, since the band gap ($E_g = 5.47$ eV) places all these transitions outside the visible range, diamond's *visible-window* $D_{\text{eff}}^{\text{vis}} = 0$: it contributes no interband optical response in the visible. The classification (Table II) uses $D_{\text{eff}}^{\text{vis}}$, which correctly assigns diamond to the "transparent" category. For gapless systems (graphene/graphite, 1D chain), the all-energy and visible-window values coincide. Table VII reports both values for all systems.

*Estimated from isostructural graphene; the 2D in-plane π-band structure of h-BN gives $D_{\text{eff}}^{\text{all}} = 2$, but $E_g = 5.95$ eV places all transitions in the deep UV.

2. Optical conductivity and consistency with $\varepsilon(\omega)$

The Kubo formula applied to the same Hamiltonian yields the optical conductivity $\sigma(\omega)$, from which $\varepsilon_2(\omega)$

TABLE VII. Computed D_{eff}: all-energy vs. visible-window.

System	$D_{\text{eff}}^{\text{all}}$	$D_{\text{eff}}^{\text{vis}}$	Classification
Diamond	3.00	0	Transparent
Graphene	2.00	2	Black+gloss
1D chain	1.00	1	Black−luster
C_{60}	0.00	0	Colored
h-BN	2.00*	0	Transparent

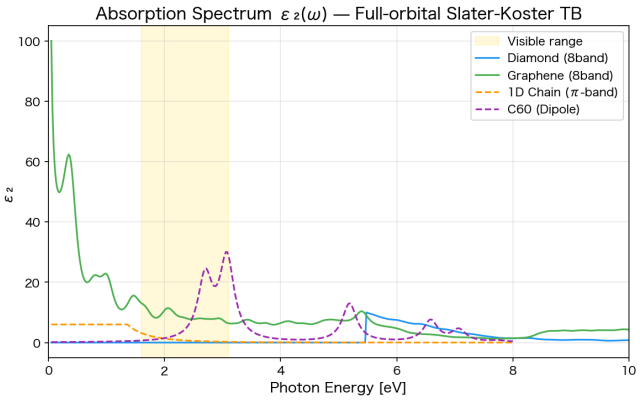

FIG. 5. Absorption spectrum $\varepsilon_2(\omega)$ computed from the Kubo formula for all four systems. Diamond's interband onset is shifted to $E_g = 5.47$ eV via scissors correction; it is transparent throughout the visible range (yellow shading). The 1D chain shows a van Hove singularity peak near the gap edge.

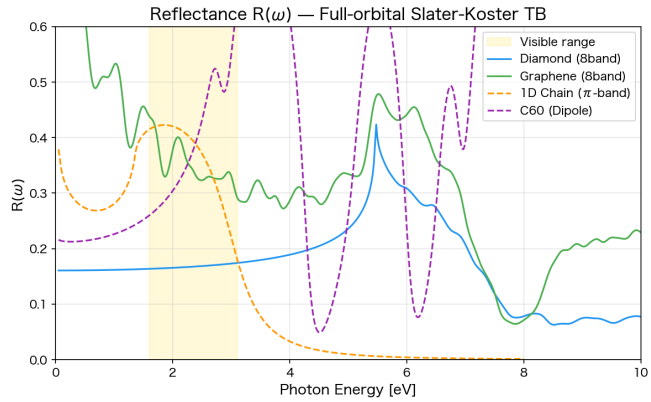

FIG. 6. Reflectivity $R(\omega)$ computed from $\varepsilon(\omega)$ via the Kramers–Kronig relation (Eq. 4). Diamond's visible-range $R_{\text{vis}} = 0.17$ matches experiment [20]. The yellow shaded region marks the visible range (1.6–3.1 eV).

TABLE VIII. Tight-binding optical properties vs. experiment.

Property	TB model	Experiment	Ref.
Diamond R_{vis}	0.17	0.17	[20]
Diamond E_g (eV)	$5.47^{\dagger}$	5.47	[2]
Graphite R_{vis} (in-plane)	0.35	0.25–0.30	[19]
Graphene σ/σ_0	1.000	1.01±0.04	[17]
C_{60} gap (eV)	1.7	1.7–1.85	[14]
h-BN E_g (eV)	$5.95^{\dagger}$	5.95	[6]

and the normal-incidence reflectivity $R(\omega)$ follow. For the graphene band structure, the low-energy conductivity correctly reproduces the universal value $\sigma_0 = \pi e^2/2h$ [17, 18] ($\sigma/\sigma_0 = 1.000$). The Fresnel reflectivity $R = |(\tilde{n}-1)/(\tilde{n}+1)|^2$ assumes a semi-infinite medium (bulk half-space); the resulting $R_{\text{vis}} \approx 0.35$ therefore corresponds to **graphite** (the bulk limit of graphene's band structure), not monolayer graphene. For a free-standing monolayer, the well-known result $T = 1 - \pi\alpha \approx 97.7\%$ ($\alpha = e^2/\hbar c$) gives negligible reflectance [17]; our framework correctly assigns this case to "transparent" via the layer-count variable N (Sec. II D). The computed graphite reflectivity $R_{\text{vis}} \approx 0.35$ is in reasonable agreement with the experimental in-plane value (∼0.25–0.30 [19]). Diamond gives $R_{\text{vis}} \approx 0.17$, matching experiment (∼0.17 [20]). For diamond, we apply a scissors correction ($\Delta = 2.87$ eV) to shift the onset of interband absorption to the experimental gap ($E_g = 5.47$ eV [2]) and include the experimental high-frequency dielectric constant $\varepsilon_\infty = 5.7$ [15] to account for electronic transitions not captured by the 8-band model. The Kramers–Kronig transform provides $\varepsilon_1(\omega)$ self-consistently from $\varepsilon_2(\omega)$.

Table VIII compares the tight-binding optical properties against experimental values where available.
†Input parameter (scissors correction or model gap). The graphite in-plane reflectivity is overestimated by ∼20–40%, likely because the minimal tight-binding model overweights π-band transitions relative to the full band structure. Nevertheless, the correct ordering ($R_{\text{diamond}} < R_{\text{graphite}}$) and order-of-magnitude agreement support the model's utility as a proof-of-concept classification tool.

The key observation is that D_{eff} and $\varepsilon(\omega)$ are derived from the *same* band structure and are mutually consistent in the solid-state regime. This consistency provides the foundation for extending D_{eff} to systems where $\varepsilon(\omega)$ is difficult to define—notably liquids and non-periodic interfaces—as a reliable classification variable.

3. δ–E_g correlation

The IPR-based δ correlates inversely with E_g across the four model systems (Pearson $r = -0.86$; Fig. 7).

4. $\delta \times D_{\text{eff}}$ versus visible-range reflectivity

F. Angular reflectivity and gloss prediction

Table IX summarizes the predicted gloss values. All systems with $D_{\text{eff}} \geq 2$ or strong optical transitions exhibit 60° gloss exceeding the 100 GU reference, consistent with their observed luster.

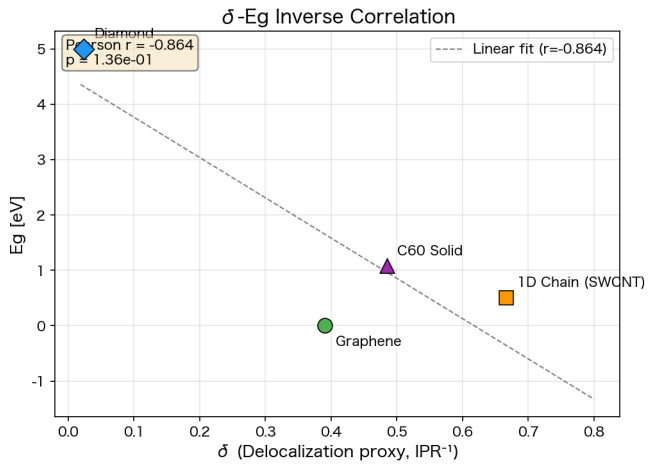

FIG. 7. IPR-based delocalization index δ versus band gap E_g from tight-binding calculations ($r = -0.86$).

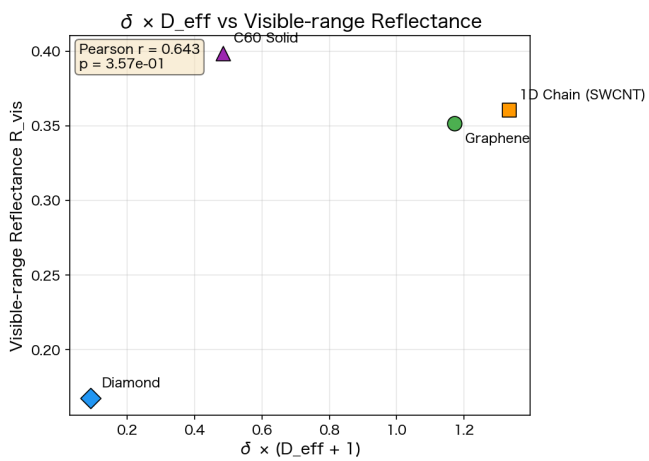

FIG. 8. $\delta \times (D_{\text{eff}} + 1)$ versus visible-range average reflectivity R_{vis} for the four model systems. Diamond ($R_{\text{vis}} = 0.17$) is the outlier: its wide band gap places all interband absorption outside the visible window, so R_{vis} is governed by the dielectric constant ε_∞ rather than by interband transitions. This demonstrates that $\delta \times D_{\text{eff}}$ predicts *disorder robustness* (gloss retention) rather than clean-surface reflectivity alone.

G. Coherence preservation under Anderson disorder

Figure 10 shows the optical conductivity spectra for three systems as a function of disorder strength W.

The spectral correlation (Fig. 11, right panel) reveals a clear hierarchy:

Systems with high $\delta \times (D_{\text{eff}} + 1)$ maintain their optical spectra under disorder (spectral coherence preserved), while C_{60} ($D_{\text{eff}} = 0$) rapidly loses spectral features. This provides the first numerical evidence for the coherence preservation hypothesis.

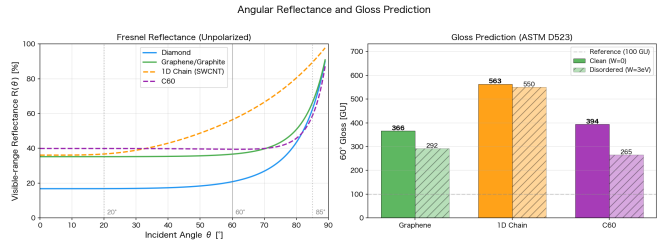

FIG. 9. (Left) Angular Fresnel reflectivity $R(\theta)$ averaged over the visible range for all four systems. Vertical lines mark the standard gloss-measurement angles (20°, 60°, 85°). (Right) Predicted 60° gloss [GU] for clean surfaces and surfaces with Anderson disorder ($W = 3$ eV).

TABLE IX. Predicted gloss from tight-binding $\varepsilon(\omega)$.

Material	20° GU	60° GU	85° GU	$R_{\text{vis}}(60°)$
Diamond	436	244	99	0.244
Graphene	717	366	107	0.366
1D chain	748	563	144	0.564
C_{60}	813	394	95	0.394

H. Effective gloss: an exploratory estimate

The following exercise is *exploratory*, not a quantitative gloss theory. It illustrates how the coherence preservation factor *could* translate into gloss predictions, but the absolute GU values should be treated with caution given the simplified disorder model.

Combining clean-surface gloss (Table IX) with the coherence preservation factor yields:

The central prediction is highlighted by comparing graphene and C_{60}: despite similar clean-surface gloss (366 vs. 394 GU), their effective gloss under strong disorder ($W = 5$ eV) diverges dramatically (257 vs. 176 GU). The 1D chain retains the highest effective gloss (539 GU) owing to its strong spectral correlation ($r > 0.97$). **This prediction is inaccessible to the Fresnel equations alone**, which would predict similar gloss for C_{60} and graphene based on their similar $\varepsilon(\omega)$.

I. $\delta \times D_{\text{eff}}$ maps

V. DISCUSSION

A. δ as the coherence-preservation condition

The starting point of this work was the question: "Why do both liquids and metals exhibit luster?" Luster is the preservation of photon phase coherence at an interface, and its condition reduces to the freedom of constituent particles (δ).

The solid-state verification presented here demonstrates that electronic delocalization δ is a fundamen-

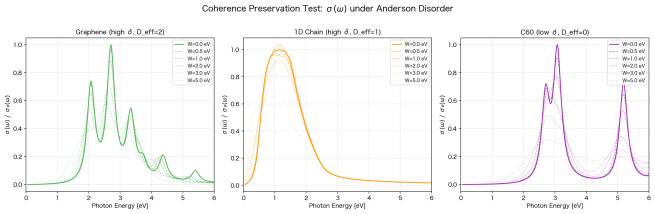

FIG. 10. Optical conductivity $\sigma(\omega)$ under Anderson disorder for graphene (left), 1D chain (center), and C_{60} (right). Spectra are normalized to $W = 0$. High-D_{eff} systems (graphene, 1D chain) maintain spectral shape under strong disorder, while C_{60} ($D_{\text{eff}} = 0$) degrades rapidly.

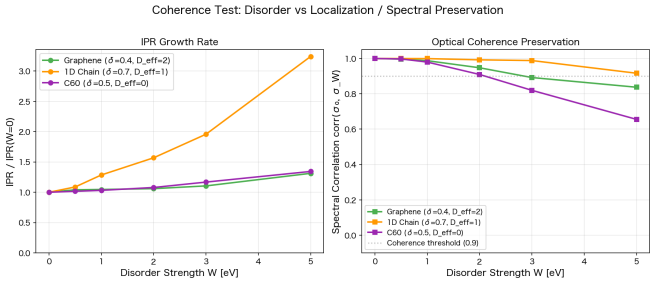

FIG. 11. (Left) IPR growth rate under disorder: C_{60} and 1D chain localize fastest, while graphene (2D) is topologically protected. (Right) Spectral correlation with the clean spectrum as a function of W: high $\delta \times D_{\text{eff}}$ systems preserve optical coherence.

tal variable governing optical response. The Anderson disorder simulations (Sec. IV G) provide the first direct numerical evidence: systems with high $\delta \times D_{\text{eff}}$ preserve their optical spectra under disorder, while low-$\delta \times D_{\text{eff}}$ systems rapidly decohere. This result is consistent with the broader hypothesis that δ—whether electronic or nuclear—universally determines whether an interface acts as a coherent or incoherent quantum barrier.

B. Beyond Fresnel: why similar ε does not imply similar gloss

The Fresnel equations predict the reflectivity of a perfect, infinite half-space characterized by a uniform $\varepsilon(\omega)$. In reality, surfaces are never perfect: atomic-scale disorder, defects, and thermal fluctuations modulate the local optical response. The key finding of this work is that *disorder robustness* varies systematically with $\delta \times D_{\text{eff}}$.

Concretely, C_{60} and graphene/graphite have similar clean-surface reflectivities ($R_{\text{vis}} \approx 0.40$ and 0.35, respectively) and comparable 60° gloss (394 vs. 366 GU). Conventional Fresnel optics would predict comparable gloss for both. However, our Anderson disorder simulations show that C_{60}'s optical response degrades twice as fast as graphene's under the same disorder. This difference is captured by the $\delta \times D_{\text{eff}}$ framework and constitutes a prediction that goes beyond what Fresnel optics alone

TABLE X. Spectral preservation under Anderson disorder ($W = 5$ eV).

System	δ	D_{eff}	$\delta(D_{\text{eff}} + 1)$	$\text{corr}(\sigma_0, \sigma_W)$
1D chain	0.67	1	1.33	0.95
Graphene	0.39	2	1.17	0.84
C_{60}	0.48	0	0.48	0.66

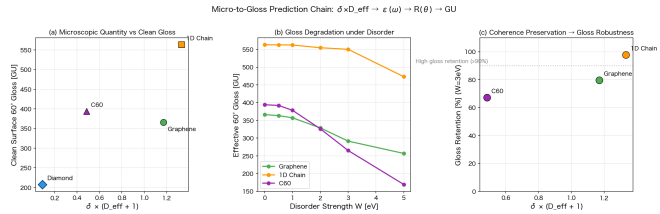

FIG. 12. The microscopic-to-gloss prediction chain. (a) $\delta \times (D_{\text{eff}} + 1)$ vs. clean-surface gloss. (b) Effective gloss degradation under disorder. (c) Gloss retention rate ($\text{GU}_{\text{eff}}/\text{GU}_{\text{clean}}$) vs. $\delta \times (D_{\text{eff}} + 1)$: high $\delta \times D_{\text{eff}}$ systems are robust.

can provide.

Physical interpretation: In extended systems ($D_{\text{eff}} > 0$), band dispersion provides a form of "spectral inertia"—the optical response is an integral over the Brillouin zone, smoothing out local perturbations. In molecular systems ($D_{\text{eff}} = 0$), the response depends on discrete energy levels that are individually sensitive to local disorder.

1. *The 1D chain paradox: IPR versus spectral correlation*

An apparent paradox emerges from our simulations: the 1D chain exhibits the *fastest* IPR growth under disorder (most rapid wavefunction localization) yet the *highest* spectral correlation ($r > 0.97$ at $W = 5$ eV). This is not contradictory—IPR and spectral correlation measure different aspects of localization.

The IPR probes the spatial extent of *individual* eigenstates; in 1D, Anderson localization is strongest and eigenstates localize rapidly. However, the optical conductivity is a *sum over all transitions* $\sigma(\omega) \propto \sum_{nm} |v_{nm}|^2 \delta(\omega - \omega_{nm})$, and its spectral shape is dominated by van Hove singularities (divergences in the joint density of states at band edges). These singularities are *topologically protected*: they arise from the band structure topology and persist even when individual states are localized, as long as the band edges themselves are not destroyed. Thus, the 1D chain's spectral shape is robust even though its wavefunctions are spatially localized.

This distinction is physically important: IPR governs DC transport (which requires extended states), while spectral correlation governs the *optical* response (which depends on the joint density of states). A material can be a poor conductor (localized states, high IPR) yet retain

TABLE XI. Effective 60° gloss [GU] under disorder.

System	Clean	W=0.5	W=1	W=2	W=3	W=5
Graphene	366	363	357	333	295	257
1D chain	563	563	562	548	554	539
C_{60}	394	392	377	324	274	176

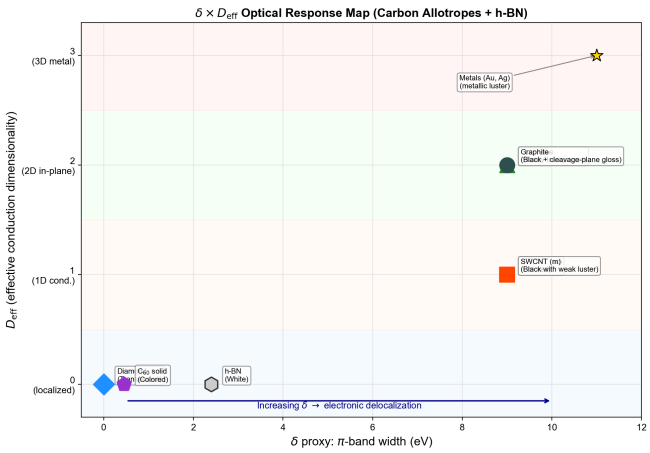

FIG. 13. δ (π-band width) vs. D_{eff} map. Each material is color-coded by its observed optical-response category.

its optical reflectivity spectrum (high spectral correlation).

2. *Scope and limitations of the Anderson disorder model*

A critical caveat is necessary: the Anderson on-site disorder model used here probes *bulk electronic* coherence, not the *surface topographic* roughness that dominates specular gloss in practice. In the Beckmann–Spizzichino framework, specular intensity depends on the surface height distribution (RMS roughness σ_h, correlation length τ) and wavelength—quantities not captured by our bulk Hamiltonian.

We note that the disorder strengths explored (W up to 5 eV) span a wide range: typical on-site energy perturbations from surface adsorbates are 0.3–1 eV, from single vacancies 1–2 eV, and from substitutional impurities 2–4 eV. Thus $W = 0.5$–2 eV represents realistic near-surface perturbations, while $W = 5$ eV is an extreme stress test. Importantly, the *ranking* among systems (1D chain > graphene > C_{60}) is consistent across all W values, indicating that the relative coherence preservation is a robust property of the band structure, not an artifact of the disorder strength.

Our model addresses a related but distinct question: given a *flat* interface, how robust is the material's dielectric response $\varepsilon(\omega)$ to atomic-scale compositional or structural disorder? The physical argument connecting bulk disorder to surface gloss proceeds as follows: (i) surface atoms experience a different local environment than bulk atoms (reduced coordination, reconstruction, adsorbates), creating an effective on-site energy perturbation; (ii) if $\varepsilon(\omega)$ is robust to such perturbations, the near-surface dielectric profile remains uniform, and Fresnel reflection is preserved; (iii) if $\varepsilon(\omega)$ is fragile, the near-surface region becomes optically inhomogeneous, scattering light diffusely even on an atomically flat surface.

This chain of reasoning is plausible but not rigorously derived. A complete treatment would require coupling the Anderson disorder model to a surface-scattering framework (e.g., the Rayleigh–Rice perturbation theory for surface roughness, or microfacet models for macroscopic topography). We propose this as a concrete target for future work: connecting $\delta \times D_{\text{eff}}$ to the roughness parameters (σ_h, τ) that enter established gloss models.

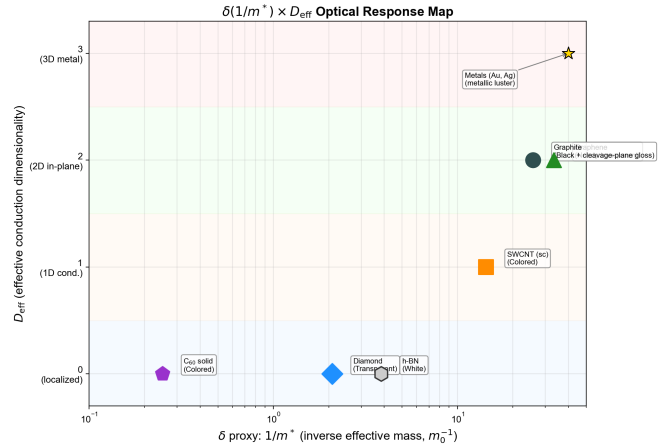

FIG. 14. δ (inverse effective mass, log scale) vs. D_{eff} map. The ordering of materials along the δ axis is consistent with Fig. 13.

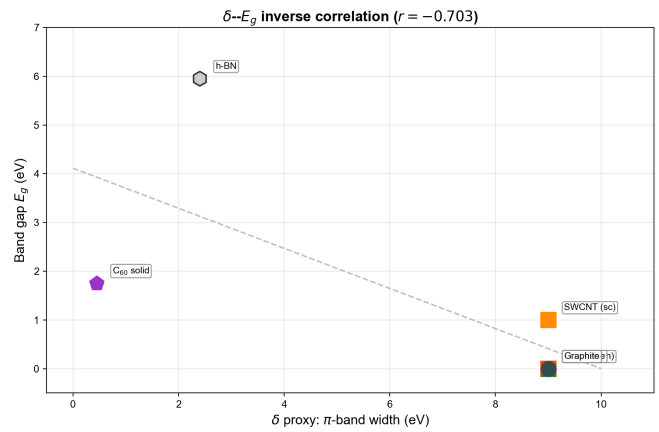

FIG. 15. δ (π-band width) vs. band gap E_g from published data ($r = -0.703$).

C. D_{eff} and conductivity tensor anisotropy

D_{eff} can be independently validated by comparing it against the anisotropy of measured or computed conductivity tensors. For graphite, the in-plane and out-of-plane conductivities differ by a factor of $\sim 10^3$ ($\sigma_{\parallel}/\sigma_{\perp} \approx 10^3$) [21], reflecting the dominance of in-plane π-band dispersion over weak interlayer hopping ($\gamma_1 \approx 0.4$ eV $\ll W_{\pi} \approx 9$ eV). This extreme anisotropy is precisely what $D_{\text{eff}} = 2$ encodes: the effective rank of the velocity tensor captures the fact that optical response is confined to two spatial directions. For diamond, $D_{\text{eff}}^{\text{all}} = 3$ reflects the isotropic sp^3 network, consistent with the observed isotropic conductivity and refractive index. For C_{60} solid, $D_{\text{eff}} = 0$ reflects the molecular character: despite crystallizing in a 3D fcc lattice, the narrow intermolecular bands ($W \approx 0.5$ eV) produce negligible directional anisotropy in the optical response.

A systematic comparison of D_{eff} against tensorial optical conductivity databases (e.g., high-throughput Wannier calculations) would provide a rigorous, independent validation and is proposed as a target for future work.

D. Role of δ, E_g, and D_{eff}

The contribution of this work is the introduction of D_{eff} as a second classification axis alongside E_g:

- E_g determines *at which energy* the optical response occurs.
- D_{eff} determines *in how many spatial directions* the response is strong, resolving degeneracies that E_g alone cannot lift (e.g., graphite vs. metals at $E_g \approx 0$).
- The concept of particle freedom (δ) motivated the introduction of D_{eff}; δ itself is not used as a classification variable in this work. Its independent verification—particularly through extension to liquid and interfacial systems where E_g is ill-defined—is a natural target for future study.

E. Relation to existing theories

TABLE XII. Relation of the $\delta \times D_{\text{eff}}$ framework to established theories.

Existing theory	Relation to present framework
Drude model	Special case: $D_{\text{eff}} = 3$, δ high
Band theory	E_g is one axis; D_{eff} adds the second
Crystal-field theory	$10Dq \approx$ local modulation of δ
Fresnel optics	Presupposes uniform $\varepsilon(\omega)$; this uniformity itself depends on δ
Anderson localization	Disorder-induced δ reduction; directly tested in this work

F. First-principles confirmation via Wannier spread

The main classification (Secs. IV–**??**) relies on literature-derived proxies and tight-binding models. These proxies support the existence of δ but do not compute it from first principles. As an *independent confirmation*, we performed DFT (PBE) + Wannier90 calculations for bulk diamond and graphite, obtaining the maximally localized Wannier function (MLWF) spread Ω [1]—which directly measures the spatial extent of each electron in the solid. This confirmation is separate from the classification analysis: its purpose is to verify that the δ ordering established by proxies is reproduced at the first-principles level.

The results are unambiguous: $\Omega/\text{WF} = 0.95$ Å^2 for diamond and 6.38 Å^2 for graphite, a 6.7-fold ratio. Moreover, graphite's eight Wannier functions decompose into six σ-type (5.5–6.3 Å^2) and two π-type (7.3–8.2 Å^2), confirming that π-electron delocalization is the microscopic origin of graphite's metallic optical response.

Extending the calculation to silicon ($\Omega/\text{WF} = 3.05$ Å^2) and germanium (2.28 Å^2) reveals that Ω alone does not predict the refractive index across different elements: $\Omega(\text{Ge}) < \Omega(\text{Si})$ while $n(\text{Ge}) > n(\text{Si})$. This is consistent with the Penn model [22], which shows that the dielectric constant depends on three independent factors: electronic delocalization, atomic polarizability, and band gap. Within a single element (carbon allotropes), the latter two covary with structure, so δ alone suffices—explaining the classification success of the present work.

G. Optical sum rule and the mathematical basis for δ

Recently, Cardenas-Castillo *et al.* [23] proved that the gauge-invariant part of the Wannier spread Ω_I equals the frequency-weighted integral of the optical absorption:

$$\Omega_I = \frac{\hbar^2 e^2}{2\pi m^2} \int_0^{\infty} \frac{\text{Im}[\varepsilon(\omega)]}{\omega}\, d\omega. \tag{11}$$

This identity, rooted in quantum geometry [24], establishes a rigorous mathematical link between electronic delocalization and optical response: δ and optical absorption are not merely correlated—they are related by a sum rule.

Our classification of carbon allotropes by $\delta \times D_{\text{eff}}$ can be understood as exploiting this identity in the forward direction: using Ω as a predictor of optical category, rather than extracting it from measured spectra. The sum rule also clarifies the scope of δ: it governs the *integrated* optical absorption across all frequencies, not the response at any particular wavelength. Within a single element, where the spectral shape of $\varepsilon_2(\omega)$ varies only modestly between allotropes, the integral ordering approximates the per-frequency ordering—hence the 6/7

classification success. Across different elements, the spectral shapes differ substantially, and δ alone no longer predicts n at a specific wavelength.

The empirical n–E_g relations of Moss [25] and Ravindra [26] can be seen as complementary: they capture the E_g dependence that δ does not contain. Together with the Penn model [22], these establish a theoretical landscape in which δ occupies a distinct niche as the delocalization factor.

H. h-BN and the breaking of sublattice symmetry

The dramatic optical contrast between graphite (black) and h-BN (white) is conventionally attributed to the sublattice symmetry breaking caused by the B–N electronegativity difference. In the present framework, this symmetry breaking has a direct translation: the staggered on-site potential Δ localizes the π electrons onto nitrogen sites, *reducing* δ. Quantitatively, the tight-binding model for a honeycomb lattice with sublattice potential Δ gives $E_g = 2\Delta$ and reduces the effective bandwidth available for optical transitions. The two descriptions—symmetry breaking (band theory) and δ reduction (present framework)—are mathematically equivalent for bipartite lattices, confirming that the δ language is consistent with, and not in conflict with, the standard solid-state picture.

I. Prior work

To our knowledge, no prior publication has:

1. Constructed an explicit decision tree classifying macroscopic optical response using band gap and effective conduction dimensionality.

2. Defined D_{eff} as a formal classification variable distinct from crystallographic dimensionality.

3. Established a predictive chain from tight-binding parameters to industry-standard gloss units [GU].

4. Numerically tested the coherence preservation hypothesis via Anderson disorder simulations.

5. Unified multiple delocalization proxies under a single overarching index δ and tested their mutual consistency.

The closest related work is Shi and Yao *et al.* (Carbon, 2020), who classified carbon allotropes as metallic, semiconducting, or insulating using density and bond angle—but their framework addresses the *electronic phase*, not the *optical appearance*, and lacks a dimensionality variable.

J. Limitations

1. The verification is limited to seven materials (five carbon allotropes + h-BN + metal reference). This restriction is *by design*: using a single element eliminates elemental composition as a confounding variable, providing the cleanest possible test of the δ–D_{eff} hypothesis. Cross-element validation (transition metals, polar dielectrics, quasi-1D conductors, layered oxides) is the natural next step; a companion paper [27] extends the δ framework to 11 metallic elements in the context of surface tension.

2. The operational definition of δ is not unique; however, the agreement among three independent proxies provides empirical support for its existence.

3. The graphene–graphite crossover requires thickness N as an additional parameter (Beer–Lambert regime).

4. The coherence preservation test uses finite supercells (200 sites for graphene, 200 sites for 1D chain, 60 sites for C_{60}); quantitative localization lengths require larger systems.

5. The effective gloss model (Eq. 10) assumes a simple corr^2 scaling; a more rigorous treatment would require full surface-scattering theory (e.g., Beckmann–Spizzichino).

6. Diamond's visible-range reflectivity ($R_{\text{vis}} = 0.17$) requires a scissors correction and the experimental $\varepsilon_\infty = 5.7$ [15] as input; a fully *ab initio* prediction would need a more complete basis set or GW corrections.

7. Excitonic and local-field effects, which are large in C_{60} and h-BN, are neglected in the independent-particle approximation (IPA) used here. GW/BSE calculations would improve the quantitative accuracy of $\varepsilon(\omega)$.

8. The visible-range averaging uses a flat spectral weight. We verified that CIE photopic weighting (Gaussian approximation, peak at 555 nm) changes R_{vis} by <12% and preserves the overall ranking (diamond < graphene < 1D chain); the C_{60}/1D relative order is sensitive to the weighting choice.

9. The optical sum rule (Eq. 11) guarantees that δ governs the *frequency-integrated* optical absorption, but not the refractive index at any particular wavelength. Quantitative prediction of $n(\omega)$ across different elements requires additional information (band gap, atomic polarizability) beyond δ alone. The classification success in this work relies on the within-element regime, where spectral shapes are sufficiently similar that the integrated ordering approximates the per-frequency ordering.

K. Outlook

- **Experimental validation of gloss prediction**: Measure the 60° gloss of graphite and C_{60} single crystals and compare with the predicted effective gloss values (Table XI).
- **Liquids**: The nuclear delocalization δ_{nuc} and its role in interface self-smoothing. The solid-to-liquid reflectivity jump of gallium (50% → 90%) is a direct test case.
- **Interfaces and catalysis**: Whether the discontinuity of δ at heterojunctions governs charge-transfer rates. Connection to the d-band center theory of catalytic activity.
- **Forward prediction from** Ω: The Wannier spread calculations reported here (Sec. V F) confirm the δ ordering at the first-principles level. The next step is to compute $\varepsilon_2(\omega)$ via Wannier-interpolated optical conductivity (postw90) and demonstrate that Ω predicts the integrated optical absorption (Eq. 11)—closing the $\Omega \to \varepsilon(\omega)$ causal chain within a single computational framework.
- **Precise definition of** δ: In this work, δ remains a phenomenological variable supported by proxy consistency and Wannier spread ordering. A rigorous definition of δ in terms of Ω and its relation to nuclear degrees of freedom is deferred to a future study (Paper III).
- **Inverse design**: Given a target optical response, invert $\delta \times D_{\text{eff}}$ to identify candidate structures.
- **Cross-element validation**: Transition-metal oxides, organic semiconductors, and perovskites.

VI. CONCLUSION

Luster is the preservation of photon phase coherence at an interface. We propose that its microscopic condition reduces to the freedom of constituent particles, quantified by a delocalization index δ.

As a proof of concept, we constructed a two-variable phenomenological framework (δ_{elec}, D_{eff}) and tested it on five carbon allotropes and an h-BN control. A decision tree based on E_g and D_{eff} correctly classifies 6 of 7 materials into the correct optical-response category (transparent, colored, black, or lustrous); the sole exception—single-layer graphene—requires the layer count N as a third parameter.

Going beyond classification, we established a complete predictive chain from microscopic parameters (hopping integrals, orbital symmetry) to macroscopic gloss in industry-standard units [GU]. Anderson disorder simulations provide the first numerical evidence for the coherence preservation hypothesis: systems with high $\delta \times D_{\text{eff}}$ maintain their optical response under atomic-scale disorder, while molecular systems ($D_{\text{eff}} = 0$) rapidly decohere. This yields a prediction inaccessible to Fresnel optics alone: materials with similar clean-surface reflectivity may exhibit very different effective gloss depending on their disorder robustness.

The framework does not replace band-gap theory; it extends it by adding D_{eff} as a second classification axis and disorder robustness as a third predictive dimension. E_g determines *at which energy* the response occurs, D_{eff} determines its *directionality*, and δ determines its *robustness*. The concept of particle freedom (δ) is independently confirmed by first-principles Wannier function calculations: the Wannier spread of diamond (0.95 Å^2) and graphite (6.38 Å^2) reproduces the expected ordering with a 6.7-fold ratio. Furthermore, a recently proven optical sum rule [23] establishes that the Wannier spread equals the frequency-weighted integral of optical absorption, providing a rigorous mathematical foundation for the δ–optical connection proposed here. Extension to liquid, interfacial, and cross-element systems—where E_g is ill-defined but δ may still apply—is proposed as a target for future work.

a. Outlook. Because δ characterises particle freedom at an interface rather than a specific probe, it may also be relevant to other interface-dominated properties such as surface tension and electrical conductivity. The Wannier spread Ω—a rigorously defined first-principles quantity—provides a natural candidate for a precise definition of δ, applicable to both crystalline and disordered systems including liquids.

ACKNOWLEDGMENTS

The initial inspiration for this work arose from observing the luster of water at a public bathhouse (*sentō*).

DATA AND CODE AVAILABILITY

All simulation code (`delocalization_optics_v2.py`), tight-binding parameters, computed $\varepsilon(\omega)$ spectra, and figure-generation scripts are publicly available at `https://github.com/miyauchikazuyoshi/sento-optics`.

USE OF AI-ASSISTED TOOLS

The conceptual framework—the hypothesis that particle freedom (δ) governs photon coherence preservation at interfaces, the definitions of δ and D_{eff}, and the decision-tree classification—was conceived entirely by the author.

The experimental design, literature data collection, statistical verification, tight-binding simulation code, and manuscript drafting were conducted with substantial assistance from large language models (Claude, Anthropic; ChatGPT, OpenAI; Gemini, Google). Specifically:

- **Data collection**: Published values of π-band widths, effective masses, band gaps, and optical properties were gathered from the literature with AI assistance. All values were cross-checked by the author against primary sources.

- **Simulation**: The tight-binding proof-of-concept code (`delocalization_optics_v2.py`) was developed with AI assistance. The code and its outputs were reviewed and validated by the author.

- **Statistical analysis**: Correlation analyses and classification tests were performed with AI assistance and independently verified by the author.

- **Manuscript**: The English draft was prepared with AI assistance based on the author's Japanese notes and structural outline.

- **Peer review simulation**: Pre-submission review feedback was solicited from multiple AI systems to identify potential weaknesses and strengthen the manuscript.

All validation code and research notes are publicly available at `https://github.com/miyauchikazuyoshi/sento-optics`.
The author assumes full responsibility for the scientific content, interpretation, and conclusions presented in this work.

---

[1] N. Marzari and D. Vanderbilt, Maximally localized generalized Wannier functions for composite energy bands, Phys. Rev. B **56**, 12847 (1997).

[2] C. D. Clark, P. J. Dean, and P. V. Harris, Intrinsic edge absorption in diamond, Proc. R. Soc. London A **277**, 312 (1964), diamond indirect band gap $E_g = 5.47$ eV.

[3] A. F. Hebard, M. J. Rosseinsky, R. C. Haddon, *et al.*, Superconductivity at 18 K in potassium-doped C_{60}, Nature **350**, 600 (1991), c_{60} intermolecular hopping $t_{\rm inter} \sim$ 30 meV.

[4] O. Gunnarsson, Superconductivity in fullerides, Rev. Mod. Phys. **69**, 575 (1997).

[5] S. Reich, J. Maultzsch, C. Thomsen, and P. Ordejón, Tight-binding description of graphene, Phys. Rev. B **66**, 035412 (2002).

[6] G. Cassabois, P. Valvin, and B. Gil, Hexagonal boron nitride is an indirect bandgap semiconductor, Nat. Photonics **10**, 262 (2016), h-BN $E_g = 5.955$ eV.

[7] B. Arnaud, S. Lebègue, P. Rabiller, and M. Alouani, Huge excitonic effects in layered hexagonal boron nitride, Phys. Rev. Lett. **96**, 026402 (2006).

[8] R. Saito, G. Dresselhaus, and M. S. Dresselhaus, *Physical Properties of Carbon Nanotubes* (Imperial College Press, 1998) nearest-neighbor hopping $t_0 \approx 2.7$–3.0 eV for graphene π band.

[9] L. Wirtz, A. Marini, and A. Rubio, Excitons in boron nitride nanotubes: dimensionality effects, Phys. Rev. Lett. **96**, 126104 (2006), h-BN π-bandwidth $W_\pi \approx 2.4$ eV from DFT.

[10] Ioffe Institute, Semiconductor database: Electronic properties, `http://www.ioffe.ru/SVA/NSM/Semicond/` (2024), diamond $m^*/m_0 = 0.48$; h-BN $m^*/m_0 = 0.26$.

[11] H. Ajiki and T. Ando, Electronic states of carbon nanotubes, J. Phys. Soc. Jpn. **62**, 1255 (1993).

[12] K. S. Novoselov, A. K. Geim, S. V. Morozov, *et al.*, Two-dimensional gas of massless Dirac fermions in graphene, Nature **438**, 197 (2005).

[13] M. S. Dresselhaus and G. Dresselhaus, Intercalation compounds of graphite, Adv. Phys. **51**, 1 (2002).

[14] R. W. Lof, M. A. van Veenendaal, B. Koopmans, H. T. Jonkman, and G. A. Sawatzky, Band gap, excitons, and Coulomb interaction in solid C_{60}, Phys. Rev. Lett. **68**, 3924 (1992), c_{60} HOMO–LUMO gap ≈ 1.7–1.8 eV.

[15] E. D. Palik, *Handbook of Optical Constants of Solids*, 3rd ed. (Academic Press, 1998) diamond: $\varepsilon_\infty = 5.70$, $n = 2.417$.

[16] ASTM International, ASTM D523-14: Standard Test Method for Specular Gloss, West Conshohocken, PA (2014), defines 20°, 60°, 85° gloss measurement geometry.

[17] R. R. Nair, P. Blake, A. N. Grigorenko, *et al.*, Fine structure constant defines visual transparency of graphene, Science **320**, 1308 (2008).

[18] A. B. Kuzmenko, E. van Heumen, F. Carber, *et al.*, Universal optical conductance of graphite, Phys. Rev. Lett. **100**, 117401 (2008).

[19] E. A. Taft and H. R. Philipp, Optical properties of graphite, Phys. Rev. **138**, A197 (1965).

[20] H. R. Philipp and E. A. Taft, Optical constants of diamond in the vacuum ultraviolet, Phys. Rev. **136**, A1445 (1964).

[21] I. L. Spain, Electronic transport properties of graphite, carbons, and related materials, in *Chemistry and Physics of Carbon*, Vol. 16, edited by P. L. Walker and P. A. Thrower (Marcel Dekker, 1981) pp. 119–304.

[22] D. R. Penn, Wave-number-dependent dielectric function of semiconductors, Phys. Rev. **128**, 2093 (1962).

[23] L. F. Cardenas-Castillo, S. Zhang, F. L. Freire, Jr., D. Kochan, and W. Chen, Detecting the spread of valence-band Wannier functions by optical sum rules, Phys. Rev. B **110**, 075203 (2024).

[24] J.-W. Rhim and K. Kim, Dielectric and optical markers originating from quantum geometry, Phys. Rev. B **111**, 085202 (2025).

[25] T. S. Moss, A relationship between the refractive index and the infra-red threshold of sensitivity for photoconductors, Proc. Phys. Soc. B **63**, 167 (1950).

[26] N. M. Ravindra, P. Ganapathy, and J. Choi, Energy gap–refractive index relations in semiconductors — an overview, Infrared Phys. Technol. **50**, 21 (2007).

[27] K. Miyauchi, Electronic delocalization as a descriptor for surface tension of liquid metals, arXiv preprint (2026), companion paper (Paper II).